East West University

Task1 Report

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Title:	A Comparative Study on Graph Construction Strategies for Hand Gesture Recognition based on sEMG Data with Graph Neural Networks.
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Abstract

This is an exploratory analysis report for the surface electromyography (sEMG) recordings from the NinaPro and a review of five related graph-neural-network (GNN) studies. This Kaggle notebook analyzed 120 recording files from 40 subjects, with 207,712,820 time samples from 12 EMG channels for each file. All of this is analyzed: descriptive statistics, quality checks, class distributions, handcrafted features, correlations, dimensionality reduction and interactive Plotly visualizations. Results show high channel and subject variation, and the need for leakage-safe and subject-independent evaluation. Since this task is not related to training GNN, no claim of model performance is made.

1. Introduction and Objectives

Surface EMG is used to recognize the gestures or to control prosthetics by measuring the electrical activity produced by the muscles during contraction. A GNN can use the EMG channels as nodes and model the relationship between the activity patterns of the channels. The properties of the signal, the distribution of the labels, dependencies between channels and between subjects need to be determined before designing such a model. The purpose of this task is to create a reproducible EDA baseline and map out research gaps of five EMG-GNN studies.

2. Dataset and Analysis Method

The provided notebook opens up and reads the NinaPro DB2 MATLAB recordings and streams each one, to avoid filling up memory. Stratified sampling of 480,000 raw rows and 16,640 selected windows are used in the analysis for visualization and feature-level analysis. At the documented 2,000 Hz sample rate, 400 samples are acquired per channel in each window of 200 ms, with 95% overlap (10 ms). The features extracted are: root mean square (RMS), mean absolute value (MAV), waveform length (WL), zero crossings (ZC) and variance (VAR) for each channel. All subjects are used in descriptive results, and graph similarity and preprocessing for evaluation on training subjects only.

A. Dataset Summary

Property	Observed description
Name/source	NinaPro DB2; supplied Kaggle copy of MATLAB files
Application area	sEMG hand gesture recognition and prosthetic control
Files / subjects	120 subject-exercise files / 40 subjects

Samples / channels	207,712,820 time samples / 12 EMG channels
Classes	49 non-rest gesture class keys, plus rest label
Features	12 raw signal channels; 5 derived features per channel (60 total)
Balance	Non-rest window max/min ratio 1.85; rest 43.9% of samples
Missing / duplicates	0 non-finite EMG values; 0 exact duplicate sampled rows

Multiple modalities are recorded at the source; the analysis of the present study focuses on the EMG modality. The signal unit and exact history of preprocessing of the downloaded version was not independently verified. The 2 kHz rate is specified in the documentation of the data set and not as an explicit sampling-rate field.

B. Descriptive Statistics

Mean, median, standard deviation, minimum, maximum, quartiles, skewness, fisher excess kurtosis and outlier indicators are provided per channel in the notebook. To not use long recordings as representative of all the subjects, both pooled-sample and subject-balanced statistics were considered. Pooled RMS (across channels): 1.05e-05 to 1.23e-04 (stored amplitude units) = 11.8. The channel skewness varies in the range of -7.68 to 2.94, showing asymmetry and heavy-tailed behavior in some channels. The median window RMS ranges from 4.47e-06 to 3.09e-05 and the median window zero crossing count is 49 to 58 per window. The quartiles and means per channel were not created in this document but rather taken from the tables that are generated in the notebook.

C. Data Quality

No NaN/Inf EMG value was found in the 2,492,553,840 channel values of the tested sample. There were no identical EMG files found, and no duplicate subject-exercise keys or exact duplicate 12-channel rows were found. There were 18 files where there was a discrepancy in the metadata; however, no required metadata variable was missing. One of 1440 file-channel combinations had near constant (standard deviation less than 1% of median channel) or zero standard deviation, or repeated more than 50% of consecutive samples. No clipping signature was found for the clipping file channel. Eighty-two percent of the outliers per channel identified using 1.5x IQR fences were in active intervals, with the remainder being statistical outliers. Furthermore, 0.92% (108 of 11,760) of non-rest segments were robust RMS outliers. These extremes might be valid muscle activity, not errors. Removal may have the effect of stifling desirable gesture information. Some suggested precautions are metadata inspection, strong scaling if relevant, and ensuring cleaning rules are in training folds.

D. Class Balance

The non-rest gesture classes are moderately imbalanced with a ratio of 1.80 (raw samples) and 1.85 (windows), while the ratio of segments is 1.00. The most common window class that isn't a rest class is E2_G34 (145,362 windows) and the least is E1_G14 (78,678 windows). The percentage of rest values are 43.9% for raw samples. A subject-by-class heatmap is included, as are class-count bar charts, including all 49 non-rest classes across all 40 subjects. The class support of overlapping windows is not independent, so class support must also be evaluated per repetition and per segment. Overall accuracy is not enough, it is better to use macro-F1, balanced accuracy (true positive rate), per-class recall and confusion matrices.

E. Feature Distributions

Histograms, boxplots and violin plots show that the magnitude of signals in the channels is different and the tails are asymmetric, and there is significant overlap between gestures. The mean absolute amplitude is 1.82e-05 for active samples, and 7.22e-06 for rest, which is roughly a factor of 2.52 greater for the active samples. For mean-channel log-RMS, the variance accounted for by class was 0.08, 0.11, and 0.05 in exercises 1, 2, and 3, respectively, and the variance accounted for by subject was 0.37, 0.42, and 0.47, respectively. This means that the amplitude differences are well confounded with the identity of the participants. While a single amplitude summary is useful, spatial activation patterns across channels may be more useful, but must be tested using a trained classifier.

F. Correlation and Redundancy

The notebook shows comparisons between Pearson correlations of raw EMG samples, correlations of RMS envelopes and Spearman correlations. The median absolute off-diagonal channel correlation for 100 raw signals is 0.007, that for the same number of RMS envelopes is 0.096 and that for the same number of the Spearman correlation is 0.038, with maximum absolute Pearson correlation of 0.176 (raw) and 0.456 (envelope) for 100 signals. There are no raw or envelope channel correlations with values greater than $|r| > 0.9$. In total there are 72 feature pairs with $|r| > 0.9$, all in the same channel of the 60 handcrafted features. Don't delete these automatically, they are candidates for redundancy analysis. Envelope-based relationships are realistic graph edges, and, however, statistical association does not necessarily denote anatomical association.

G. PCA, t-SNE, UMAP and 3D Visualization

The notebook generates two representations: the flattened raw windows (4,800 values) and handcrafted feature vectors (60 values). The first three training-fitted PCA components account for 3.81%, 2.20% and 1.71% of the variance in raw-window features, compared to the handcrafted-feature shares of 38.67%, 10.38%, and 8.09%. In exercise (1): Two dimensional PCA class silhouette scores are: -0.18 (handcrafted); -0.15 (raw). For exercise 1, t-SNE class silhouettes are -0.15 (handcrafted) and -0.11 (raw); UMAP scores are -0.12 and -0.11. Visually, these low dimensional projections appear to not cleanly separate gestures with negative scores, indicating that

these projections may not distinguish gestures effectively. A 3D PCA preview was also created. Such classification conclusions can not be drawn directly due to projection artifacts and sampling selections.

H. Interactive Visualizations

Eleven standalone Plotly HTML visualizations were generated by the notebook: multichannel raw time series, amplitude distributions, class counts, subject-channel statistics, channel correlations, PCA by class, PCA by subject, 3D PCA by class, classwise features, graph topology comparison and graph threshold sweep. They are accompanied by a title and short reading. The interactive outputs support hovering and zooming and filtering; the notebook containing the creation of these outputs was uploaded but the stand-alone HTML files are not included in this report. The notebook should be run by the reader, or the output files from the notebook should be downloaded from Kaggle.

3. Implications for graph construction and evaluation

The analysis suggests a 12 nodes, each corresponding to an EMG channel. There are 66 edges in a fully connected undirected graph; undirected graphs with undirected RMS envelope correlation KNNs are 19, 26, 35, 41 and 47 edges for K=2, 3, 4, 5 and 6 respectively. The absolute-correlation thresholds are 0.1, 0.3 and 0.5, yielding 39, 8 and 0 edges, respectively. The structure of the graph, then, should be regarded as an experimental option. Five-fold subject-wise evaluation: The notebook specifies that there should be 28 training, 4 validation and 8 test subjects per fold and no overlap between subjects and source segments across the different partitions. The only part of each fold to be used for scaling, edge estimation and tuning should be the training part.

4. Related Work : Six EMG-GNN Studies

Study	Approach and reported	Main limitations
Recognizing hand gestures using cross-user electromyography. Xu et al., 2024: Cross-User Electromyography Pattern Recognition	CNN plus multi-view spatial-temporal GCN; 96.9% within-subject; 68.0% user-independent; 92.3% with the transfer learning.	Cross-user generalization is still a challenge; motivates subject independent tests.
Lee et al. (2025) presents a total support effort of 2.26x104.12 N on the chair.Lee s. et al. (2025) reported total support of 2.26x104.12 N on the chair.	Compared explicit general-purpose sensor graphics and temporal edges, achieving accuracy of reported >99.3% intra-subject and up to 92.4% inter-subject in DB5.	Possible overlapping-window leakage; drives guided comparison of graphs.

Hu et al. (2024), A Novel Multi-Feature Fusion Network	A Pearson correlation graph using spatial partitions and temporal, frequency fusion reported results of 98.78% for MyoData and 91.60% for DB5.	Gains are unclear due to subject dependent assessment/s and complex fusion.
Hand Gesture Recognition based on Surface Electromyography Signals (Surban Electromyography Signals) (2025) is an article written by Zhou et al. that appeared in IEEE Transactions on Neural Systems and Rehabilitation (2025).	Covariance-refined GCN on NinaPro DB2; reported 88.02% (exercise B), 81.30% (C), 93.30% (D).	Seen as the most directly comparable to DB2 work, still open are unseen-subject and alternative graph definitions.
Massa et al. (2022), Graph Neural Networks for HD EMG-based Movement Intention Recognition	GraphSAGE on 128channeled HD-EMG achieved a mean classification error of 8.75%.	Different sensor density from DB2, limited baseline and cross subject comparisons.
Lee et al. (2025), explicit vs general-purpose sensor graphs.	Compared sensor graphs and temporal edges; >99.3% intra-subject and up to 92.4% inter-subject, on DB5.	May have overlapping window leakage; leads to comparison of graphs.

Above paper descriptions and performance values are copied from the research-gap table that was supplied, and were not necessarily independently verified in full publications. These scores represent "contextual" benchmarks that are not exact comparisons to each other.

5.Proposed Work direction and research gap:

Four trends are identified across the 5 studies: limited subject-independent generalization; potential positive bias from random window splitting where there is overlap in subject content; too few controlled comparisons of the construction of the sensor graph; and too much model complexity is used to hide the effect of the graph. These questions can be explored systematically in the public, 12-channel setting of NinaPro DB2. Proposed Follow-on study: Examine the nature of these fully connected and compare to the two KNN and correlation threshold graphs with the same window, using the same temporal encoders and strong CNN and LSTM baselines on the subject-wise folds. These experiments are given as suggestions, not as completed experiments in task 1.

6. Conclusion:

In Task 1 a detailed picture of the DB2 recordings of the NinaPro was created. There are no records of non-finite EMG values, some repetitive near-constant channel cases, moderate imbalance of data classes within the non-rest class and high prevalence of data classes within the rest class. The ownership of the pictures is hard to achieve: Within the picture, there is much variation, and at the low dimensionalities of the pictures (on which there are platforms, boats and boats still in various degrees of vertical compression), there is some overlap of features that are also provided with correlated features at the perimeter of the pictures by hand. Hence, the following steps to achieve RMS envelope graph construction and disjoint checking of this graph with the subject are considered justified. The outcome is used to show illustrative evidence and is also a repeatable evaluation plan to show the improvement in performance of GNN.